

Patch: Paper 34, product table in the discriminant computation

The source page has a displayed product table in the proof computing the discriminant of a matrix ring. The table is:

		$\overbrace{c_{11} \cdots c_{1n}}^{r_1}$	$\overbrace{c_{21} \cdots c_{2n}}^{r_2}$	$\cdots$
c_{11}	$\left. \begin{array}{c} \vdots \\ c_{n1} \end{array} \right\} t_1$	(c_{ik})	0	0
$\vdots$				
c_{n1}				
c_{12}	$\left. \begin{array}{c} \vdots \\ c_{n2} \end{array} \right\} t_2$	0	(c_{ik})	0
$\vdots$				
c_{n2}				
$\vdots$		0	0	(c_{ik})

and therefore the discriminant is

$$D = \left| \begin{array}{cccc} \text{Sp}(c_{11}) & 0 & \cdots & 0 \\ 0 & \text{Sp}(c_{22}) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \text{Sp}(c_{nn}) \end{array} \right|^n = \begin{cases} e, & \text{for reduced traces,} \\ n^{n^2}e, & \text{for principal traces.} \end{cases}$$